United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final Exam Year: 2026 Trimester: Summer

CSE 4165/CSE 465 Web Programming

Total Marks: 30 Duration: 2 Hours

SET - A
09:00 AM - 11:00 AM

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. Create a JavaScript application where a user can add items to a shopping cart and see the bill update after every item is added. [10]

Index.html:

Create three input fields for **Item Name**, **Price (BDT)**, and **Quantity**, an **"Add to Cart"** button, and below it a label field to display the cart summary.

Script.js:

1. Take the **item name**, **price**, and **quantity** from the input fields.
2. Inside a function:
   a) If the item name is empty, the price is not greater than 0, or the quantity is not a positive whole number, display "Invalid item!" and do not add the item.
   b) Store each item name in an array.
   c) Add (Price × Quantity) to a running subtotal, and add the quantity to the total quantity.
   d) Calculate the discount on the subtotal:
      - 5000 BDT or more → 10%
      - 2000 BDT or more but less than 5000 BDT → 5%
      - Less than 2000 BDT → no discount
   e) Calculate **VAT = 5% of (Subtotal − Discount)**.
   f) Calculate **Grand Total = Subtotal − Discount + VAT**.
   g) Display the items, total quantity, subtotal, discount, VAT, and grand total (rounded to 2 decimal places).
   h) If the total quantity is more than 20, also display **"Bulk order! Free delivery."**

Sample Input (Item,Price and Quantity, entered in order)	Sample Output
Pen 20 3, Notebook 80 5	Items: [Pen, Notebook], Qty: 8, Subtotal: 460.00, Discount: 0.00, VAT: 23.00, Grand Total: 483.00
Bag 1500 1, Shoes 2200 1	Items: [Bag, Shoes], Qty: 2, Subtotal: 3700.00, Discount: 185.00, VAT: 175.75, Grand Total: 3690.75
Pencil 10 15, Eraser 5 10, Calculator 1200 4	Items: [Pencil, Eraser, Calculator], Qty: 29, Subtotal: 5000.00, Discount: 500.00, VAT: 225.00, Grand Total: 4725.00, Bulk order! Free delivery.
Pen -20 2	Invalid item!

2. Write a PHP program for a hotel booking bill calculator. The program should take four inputs from the user: **room type**, **number of weekday nights**, **number of weekend nights**, and **number of guests**. [10]
Room type should be either ***Standard***, ***Deluxe***, or ***Suite***. The **weekday rate per night** depends on the room type:

- Standard: 3,000 Tk
- Deluxe: 5,000 Tk
- Suite: 8,000 Tk

Room Charge: The weekend rate is 25% higher than the weekday rate. (e.g. For a Deluxe room, the weekend rate is: 5000+5000*0.25=6,250 Tk per night.)

Room charge = (weekday nights × weekday rate) + (weekend nights × weekend rate)

Long-Stay Discount: Based on the total number of nights (weekday + weekend), **applied to the room charge only**:

- Less than 3 nights: no discount
- 3 to 6 nights: 5% discount
- 7 nights or more: 10% discount

Extra Guest Charge: Each room includes **2 guests at no extra cost**. Every additional guest is charged 800 Tk per night for the total number of nights. (e.g. 4 guests for 5 nights = 2 extra guests × 800 × 5 = 8,000 Tk.)

Bill Calculation:

Subtotal = room charge after discount + extra guest charge

Service Charge & VAT: A combined 15% of the subtotal.

Final Bill= subtotal + service charge & VAT

The program should display the room charge, discount amount, extra guest charge, service charge & VAT amount, and the final payable bill.

HTML Form	PHP Output
Room Type: Deluxe Weekday Nights: 4 Weekend Nights: 2 Number of Guests: 4 Calculate	Room Charge: 32500 Tk Discount: 1625 Tk Extra Guest Charge: 9600 Tk Service Charge & VAT: 6071.25 Tk Final Bill: 46546.25 Tk
Room Type: Standard Weekday Nights: 5 Weekend Nights: 0 Number of Guests: 2 Calculate	Room Charge: 15000 Tk Discount: 750 Tk Extra Guest Charge: 0 Tk Service Charge & VAT: 2137.5 Tk Final Bill: 16387.5 Tk

3. *Create a database named* '**bd_cricket**' and a table named '**match_performance**' with the following structure and values: [10]

Id	PlayerName	Team	Venue	Runs	Balls
1	Shanto	DHK	Mirpur	112	98
2	Shanto	DHK	Sylhet	45	52
3	Litton	KHL	Sylhet	88	70
4	Litton	KHL	Mirpur	67	60
5	Miraz	CTG	Khulna	30	41
6	Taskin	CTG	Khulna	52	44

Now, do the following (write full PHP–MySQL code):

1. For **each venue**, display the **venue name**, the **number of performances** and the **average runs**, but only include venues **whose average runs are above 50**.
2. For **every performance** at the venue **"Mirpur"**, increase the runs by **5%** as a home-ground adjustment.
3. Display the **top 3** performances by **strike rate (Runs / Balls × 100)**, showing the **player name**, **venue** and **strike rate**.
4. Display each performance's **player name** and **team**, with the label **"Above Par"** if its runs are **above the average runs of its own team**, otherwise **"Below Par"**.